



INSTITUTE OF AERONAUTICAL ENGINEERING (Autonomous)

Dundigal, Hyderabad - 500 043

QUESTION BANK

Department	Information Technology				
Course Title	COMPUTER SYSTEM ARCHITECTURE				
Course Code	AITE01				
Program	B. Tech				
Semester	III	CSE / IT / CSE(AI&ML) / CSE(DS)			
Course Type	Core				
Regulation	BT-25				
Course Structure	Theory			Practical	
	Lecture	Tutorials	Credits	Laboratory	Credits
	3	0	3	-	-
Course Coordinator	Dr. P L Srinivasa Murthy, Professor				

COURSE OBJECTIVES:

The students will try to learn:

I	The concepts of register transfer logic and arithmetic operations, instruction format, and instruction cycle.
II	The basic components of computer systems, functionality, and interactions with the components.
III	Memory hierarchy, memory management and I/O management.
IV	Pipelining and Multiprocessor techniques for the improvement of efficiency.

COURSE OUTCOMES:

After successful completion of the course, students should be able to:

CO 1	Explain Boolean algebra, digital logic circuits, and data representation techniques used in digital computers.	Understand
CO 2	Analyze and design combinational and sequential logic circuits using standard design procedures and HDL concepts.	Analyze
CO 3	Apply register transfer language (RTL) and micro-operations to describe data flow and instruction execution in a basic computer system.	Apply
CO 4	Interpret instruction formats, addressing modes, and control unit operations in CPU organization and design.	Understand
CO 5	Analyze computer arithmetic algorithms and evaluate their role in processor performance and accuracy.	Analyze

CO 6	Explain memory hierarchy, cache organization, and input–output mechanisms including interrupts and DMA.	Understand
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QUESTION BANK:

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MODULE I				
INTRODUCTION TO DIGITAL LOGIC AND COMPUTER ARCHITECTURE				
PART A-PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				
1	Explain the fundamental laws of Boolean algebra. Simplify the Boolean expression $(A+B)(A+B')+AB$ using Boolean theorems.	Understand	This would require the learner recalls the definition and explains register-to-register transfer using control signals.	CO 1
2	A digital security system requires a logic circuit that activates an alarm only when two specific sensors are ON. Design the circuit using logic gates and explain its operation.	Understand	This would require the learner interprets bus structures and explains how data moves with control functions.	CO 1
3	Convert the decimal number 245 into Binary, Octal, and Hexadecimal. Compare the advantages of each number system.	Understand	This would require the learner applies knowledge of microoperations to implement logical designs.	CO 1
4	Represent –37 using 1's complement and 2's complement in 8-bit format. Explain why 2's complement is preferred.	Understand	This would require the learner explains the concept of logicmicrooperations and interprets examples.	CO 1
5	Convert the decimal number 53.625 into fixed-point and IEEE-754 single-precision floating-point representations.	Understand	This would require the learner applies shifting techniques and derives the output for given binary input.	CO 1

6	Compare fixed-point and floating-point representations. Determine which representation is more suitable for scientific computing with justification.	Understand	This would require the learner breaks down ALU structure and explains how it integrates multiple operations.	CO 1
7	Design a 3-to-8 Decoder using logic gates and explain its truth table and applications.	Apply	This would require the learner applies binary arithmetic rules to solve examples using complement techniques.	CO 1
8	Design an 8-to-3 Encoder and explain how it converts multiple input lines into binary output.	Understand	This would require the learner recalls IEEE 754 format and explains floating-point operations with example.	CO 2
9	Design a 4-to-1 Multiplexer using logic gates and explain how it can implement Boolean functions.	Apply	This would require the learner applies BCD arithmetic rules to perform decimal addition step by step.	CO 1
10	Explain the operation of a 1-to-4 De-Multiplexer with a neat logic diagram and truth table.	Understand	This would require the learner recalls and explains the basic arithmetic rules applied in floating-point math. .	CO 1
PART-B LONG ANSWER QUESTIONS				
1	Explain Boolean algebra and its postulates. Simplify complex Boolean expressions using Boolean laws with suitable examples.	Understand	This would require the learner recalls the definition and explains register-to-register transfer using control signals.	CO 1
2	Discuss binary codes and number systems. Convert between Decimal, Binary, Octal, and Hexadecimal with examples.	Understand	This would require the learner interprets control signal behavior and illustrates it with system-level visualization.	CO 1

3	Explain Binary Storage and Registers. Describe different types of registers and their applications in digital systems.	Understand	This would require the learner describes the system organization and control mechanisms in bus-based architecture.	CO 1
4	Explain logic gates in detail. Draw symbols, truth tables, Boolean expressions, and practical applications of each gate.	Understand	This would require the learner designs a small-scale system and applies logic to demonstrate control-based transfer.	CO 1
5	Discuss various data types used in computer systems. Explain their significance in data representation.	Understand	This would require the learner applies arithmetic operations and shows how they are implemented at microoperation level.	CO 1
6	Explain complements in digital systems. Illustrate the formation of 1's and 2's complement and their applications in subtraction.	Understand	This would require the learner constructs an adder and explains internal logic for micro-level computation.	CO 1
7	Explain Fixed Point Representation and Floating Point Representation. Compare their advantages, disadvantages, and applications.	Understand	This would require the learner explains Boolean logic operations with visual tables and diagrams.	CO 1
8	Describe the IEEE-754 Floating Point Standard. Explain normalization, exponent, mantissa, and bias with examples.	Apply	This would require the learner designs a functional multiplexer-based circuit handling multiple logic operations.	CO 2
9	Explain the working of Decoders. Design a 3-to-8 Decoder and discuss its practical applications.	Understand	This would require the learner explains and compares various shift operations using binary examples.	CO 1
10	Explain the operation of Encoders. Design an 8-to-3 Encoder with truth table and logic diagram.	Apply	This would require the learner constructs a circuit using multiplexers to perform parallel shift operations.	CO 1

11	Explain Multiplexers in detail. Design a 4-to-1 Multiplexer and implement a Boolean function using it.	Understand	This would require the learner analyzes ALSU structure and its integration of arithmetic, logic, and shift functions.	CO 1
12	Explain the working of De-Multiplexers. Design a 1-to-8 De-Multiplexer and discuss its applications.	Apply	This would require the learner constructs and demonstrates the functionality of a multi-operation control unit.	CO 1
13	Compare Decoder, Encoder, Multiplexer, and De-Multiplexer with suitable examples.	Apply	This would require the learner solves binary arithmetic using 1's and 2's complement examples.	CO 1
14	Explain the working principles of SR, JK, D, and T Flip-Flops with excitation and characteristic tables.	Understand	This would require the learner compares number formats and evaluates overflow scenarios.	CO 1
15	Design a synchronous circuit using JK Flip-Flops to generate a specified sequence.	Understand	This would require the learner explains IEEE 754 standard and demonstrates arithmetic operations.	CO 1
16	Explain Shift Registers. Compare Serial and Parallel Shift Registers with suitable diagrams.	Understand	This would require the learner performs a step-by-step arithmetic solution using floating-point normalization	CO 1
17	Explain Counters. Compare Asynchronous and Synchronous Counters with timing diagrams.	Understand	This would require the learner applies BCD addition rules and uses correction logic to validate results.	CO 1
18	Design a MOD-10 Counter using JK Flip-Flops and explain its operation.	Understand	This would require the learner constructs and explains BCD arithmetic with carry correction.	CO 1

19	Explain how Flip-Flops, Shift Registers, and Counters are used in modern computer systems and communication devices.	Understand	This would require the learner distinguishes operations and analyzes control sequencing mechanisms.	CO 1
20	Design a digital system combining Logic Gates, Multiplexers, Flip-Flops, and Counters to perform a real-time digital application. Explain its operation with neat diagrams.	Understand	This would require the learner evaluates the role of microoperations in enhancing CPU functionality.	CO 1
PART-C SHORT ANSWER QUESTIONS				
1	Define Boolean Algebra.	Remember	-	CO 1
2	State De Morgan's Theorems.	Remember	-	CO 1
3	What is a Binary Code?	Understand	This would require the learner to recall arithmetic operations then explain the micro-operations	CO 1
4	Define Binary Storage.	Remember	-	CO 1
5	What is a Register?	Understand	This would require the learner to recall shift operations then explain the shift operations	CO 1
6	Define Logic Gates.	Remember	-	CO 1
7	What is the difference between AND and OR gates?	Understand	This would require the learner to recall memory then understand the tri state buffer and also write its applications	CO 1
8	Define Fixed Point Representation.	Remember	-	CO 1
9	Define Floating Point Representation.	Remember	-	CO 1
10	What is 2's Complement?	Remember	-	CO 1
10	What is 2's Complement?	Remember	-	CO 1
11	Define Decoder.	Understand	This would require the learner to define instruction then list the instruction format	CO 1

12	Define Encoder.	Understand	This would require the learner to know micro-operation then understands micro program	CO 1
13	What is a Multiplexer?	Remember	-	CO 1
14	What is a De-Multiplexer?	Remember	-	CO 1
15	Define Flip-Flop.	Remember	-	CO 1
16	List the types of Flip-Flops.	Remember	-	CO 1
17	Define Shift Register.	Remember	-	CO1
18	What is a Counter?	Remember	-	CO 1
19	Differentiate between Ripple Counter and Synchronous Counter.	Understand	This would require the learner to know micro-operation then explain the operation of micro programmed control unit	CO 1
20	List any four applications of Multiplexers in digital systems.	Remember	-	CO 1

MODULE II

INTRODUCTION TO COMPUTER ORGANIZATION

PART-A PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS

1	Explain the functional organization of a basic computer system. Illustrate the interaction among the CPU, memory, and I/O units during program execution.	Understand	This would require the learner explains instruction format structure and interprets the function of opcode and operand with examples.	CO 2
2	A computer executes an instruction that requires fetching data from memory, performing arithmetic, and storing the result back in memory. Explain each step involved in the execution cycle.	Understand	This would require the learner identifies various registers and explains their specific purpose in instruction cycles with practical examples.	CO 2
3	Compare the Von Neumann and Harvard architectures. Analyze which architecture is more suitable for embedded systems with proper justification.	Apply	This would require the learner explains RTL representation and interprets how data is transferred between registers for given instruction.	CO 2

4	Design the block diagram of a basic CPU and explain the functions of the Arithmetic Logic Unit (ALU), Control Unit (CU), Program Counter (PC), and Instruction Register (IR).	Understand	This would require the learner recalls and interprets performance metrics such as MIPS, CPI, and clock speed to explain CPU efficiency.	CO 2
5	Explain how the CPU communicates with the memory subsystem through address, data, and control buses during a memory read operation.	Apply	This would require the learner interprets the interaction of control signals during memory read and explains it using timing diagram.	CO 2
6	A computer has RAM, ROM, and Cache memory. Analyze the role of each memory type and explain how they improve overall system performance.	Understand	This would require the learner explains fetch, decode, execute phases and how each stage influences CPU operations.	CO 2
7	Explain the organization of a memory subsystem. Describe how memory interfacing is performed between the CPU and memory modules.	Remember	This would require the learner recalls the working of both I/O types and summarizes interrupt handling sequence using diagram.	CO 2
8	A processor needs to communicate with multiple input/output devices. Explain how an I/O subsystem is organized and how interfacing enables communication.	Apply	This would require the learner interprets instruction format structure and explains decoding process with rationale for field allocation.	CO 2
9	Compare programmed I/O, interrupt-driven I/O, and DMA-based I/O. Analyze which technique provides better performance for high-speed devices.	Understand	This would require the learner explains various addressing techniques with examples and evaluates their benefit in flexible coding.	CO 2
10	Explain the architecture of a simple computer with a neat block diagram. Describe the function of each hardware component.	Understand	This would require the learner explains differences in function and effect of instruction types using register-level examples.	CO 2
PART-B LONG ANSWER QUESTIONS				

1	Explain the organization of a basic computer system with a neat block diagram. Discuss the functions of all major hardware components.	Understand	This would require the learner interprets conditional execution and explains instruction flow within control unit using an example. .	CO 2
2	Explain CPU organization in detail. Describe the architecture and functions of the ALU, Control Unit, Registers, Program Counter, and Instruction Register.	Understand	This would require the learner recalls basic instruction structure and explains decoding logic with binary examples.	CO 2
3	Discuss the organization of the memory subsystem. Explain different memory components and memory interfacing techniques.	Understand	This would require the learner explains the structure and performance effect of instruction components in CPU processing.	CO 2
4	Explain memory read and memory write operations with suitable timing diagrams.	Understand	TThis would require the learner describes register types and interprets functional role through visual and RTL explanation.	CO 2
5	Explain the organization and interfacing of input/output subsystems. Discuss different methods of data transfer between CPU and I/O devices.	Understand	This would require the learner explains the role of accumulator and register usage in data operations and CPU architecture.	CO 2
6	Compare Programmed I/O, Interrupt-driven I/O, and Direct Memory Access (DMA) with suitable examples.	Understand	This would require the learner categorizes instruction types and explains their purpose using suitable examples.	CO 2
7	Explain the architecture of a simple computer and discuss the interaction among CPU, memory, and I/O devices.	Understand	This would require the learner interprets instruction lifecycle using RTL description to demonstrate control and data flow.	CO 2

8	Explain different levels of programming languages. Compare Machine Language, Assembly Language, and High-Level Languages.	Understand	This would require the learner compares two control unit types and evaluates their design strengths and weaknesses.	CO 2
9	Explain assembly language instructions with suitable examples. Discuss arithmetic, logical, data transfer, and branching instructions.	Apply	This would require the learner demonstrates timing of memory access and interprets control signal roles through diagrams.	CO 2
10	Explain the functions of an assembler and the steps involved in translating assembly language into machine language.	Remember	This would require the learner recalls the instruction cycle and describes sequential operations in CPU execution.	CO 2
11	Discuss the design principles of Instruction Set Architecture (ISA). Explain the factors influencing ISA design.	Understand	This would require the learner explains standard instruction cycle behavior and how control shifts under interrupts.	CO 2
12	Explain different instruction formats used in ISA design with suitable examples.	Understand	This would require the learner compares I/O types and explains their working principles and differences in performance.	CO 2
13	Discuss addressing modes in instruction set architecture. Explain their advantages and applications.	Understand	This would require the learner recalls and identifies how interrupts are managed using control vectors and diagrams.	CO 2
14	Compare RISC and CISC architectures with suitable examples and applications.	Understand	This would require the learner interprets instruction format types and demonstrates each with valid use cases.	CO 2
15	Explain a simple Instruction Set Architecture and describe its instruction execution process.	Understand	This would require the learner designs and explains a new instruction format and its decoding logic.	CO 2

16	Explain the fetch–decode–execute cycle in a simple computer with a neat diagram.	Understand	This would require the learner explains addressing types and provides binary-coded illustrations of each.	CO 2
17	Explain the role of buses in CPU, memory, and I/O communication. Discuss Address Bus, Data Bus, and Control Bus.	Understand	This would require the learner evaluates addressing techniques and shows how they enhance instruction flexibility.	CO 2
18	Describe the organization of registers in a CPU and explain their role during instruction execution.	Understand	This would require the learner explains and contrasts instruction types using practical CPU operation examples.	CO 2
19	Explain how memory organization and I/O organization influence computer system performance.	Understand	This would require the learner translates logic operations into RTL format and shows how data is manipulated.	CO 2
20	Design a simple computer architecture capable of executing arithmetic, logical, memory, and branching operations. Explain the working of the system with a neat block diagram.	Understand	This would require the learner recalls control flow principles and demonstrates flow with basic code structure.	CO 2
PART-C SHORT ANSWER QUESTIONS				
1	Define Computer Organization.	Remember	-	CO 2
2	List the major functional units of a computer.	Remember	-	CO 2
3	What is CPU Organization?	Remember	-	CO 2
4	Define Arithmetic Logic Unit (ALU).	Remember	-	CO 2
5	What is the function of the Control Unit?	Remember	-	CO 2
6	Define Memory Subsystem.	Remember	-	CO 2
7	What is Memory Interfacing?	Remember	-	CO 1
8	Define Input/Output Interfacing.	Remember	-	CO 2

9	What is Programmed I/O?	Understand	This would require the learner to recall the memory then list the different memory chips	CO 2
10	What is Interrupt-driven I/O?	Remember	-	CO 2
11	Define Direct Memory Access (DMA).	Remember	-	CO 2
12	What is a Simple Computer?	Understand	This would require the learner to define instruction then show the instruction format	CO 2
13	List the levels of programming languages.	Remember	-	CO 2
14	Define Machine Language.	Understand	This would require the learner to define concept of instructions then relate data transfer instructions	CO 2
15	Define Assembly Language.	Understand	This would require the learner to define number system then explain number systems in computer organization	CO 2
16	What is an Assembler?	Understand	-	CO 2
17	Define Instruction Set Architecture (ISA).	Remember	-	CO 2
18	What is an Instruction Format?	Remember	-	CO 2
19	Differentiate between RISC and CISC.	Remember	-	CO 2
20	List the advantages of Assembly Language over Machine Language.	Understand	This would require the learner to define instruction then explain the instruction set	CO 2
MODULE III				
ORGANIZATION OF A COMPUTER				
PART A-PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				

1	Explain Register Transfer Language (RTL). Write RTL statements to perform data transfer among registers R1, R2, and R3.	Apply	–	CO 3
2	A processor contains four registers. Demonstrate how register transfer operations are performed using RTL notation with suitable examples.	Understand	This would require the learner to recall number system then explain the conversion	CO 3
3	Explain the organization of a common bus system. Illustrate how data is transferred among multiple registers using a common bus.	Evaluate	This would require the learner to recall arithmetic operations then demonstrate the restoring division.	CO 3
4	A CPU must transfer data from memory location 2500H to register R1 and then to register R2. Write the sequence of RTL statements and explain each operation.	Understand	This would require the learner to recall arithmetic operation then explain the signed- magnitude operations.	CO 3
5	Design a bus transfer mechanism using multiplexers and explain its operation with a neat block diagram.	Understand	–	CO 3
6	Explain arithmetic micro-operations. Demonstrate how addition, subtraction, increment, and decrement operations are performed on register contents.	Understand	This would require the learner to recall floating points then solve problem using subtraction	CO 3

7	A processor contains two registers, A and B. Explain the sequence of arithmetic micro-operations required to compute $A \leftarrow A + B$ and $B \leftarrow B - 1$.	Understand	This would require the learner to recall arithmetic operations then explain the different arithmetic micro-operations	CO 3
8	Explain logic micro-operations. Demonstrate the execution of AND, OR, XOR, and Complement operations using register values.	Analyze	–	CO 3
9	A control unit receives two register values. Explain how logic micro-operations can be used to manipulate the data for decision making.	Understand	This would require the learner to recall interrupt cycle then related to basic computer	CO 3
10	Explain shift micro-operations. Compare logical, arithmetic, and circular shift operations with suitable examples.	Analyze	The learner to define instruction format and describe about addressing instruction then compare and analyse the direct and indirect addressing instructions	CO 3
PART-B LONG ANSWER QUESTIONS				
1	Explain Register Transfer Language (RTL). Discuss its syntax, symbols, and applications with suitable examples.	Understand	This would require the learner to recall the memory then demonstrate the memory reference instructions.	CO 4
2	Explain register transfer operations. Describe various methods of transferring data among processor registers.	Understand	The learner to define interrupt cycle and models the flow chart.	CO 3
3	Discuss the organization of a common bus system. Explain bus transfer and memory transfer operations with neat diagrams.	Apply	-	CO 3

4	Explain memory transfer operations. Discuss memory read and memory write cycles using RTL notation.	Understand		CO 3
5	Explain arithmetic micro-operations in detail. Discuss addition, subtraction, increment, decrement, and arithmetic shift operations with examples.	Understand	This would require the learner to recall the number system then translate the conversation of binary system	CO 3
6	Explain logic micro-operations. Discuss AND, OR, XOR, NOT, and selective set, clear, and complement operations.	Understand	This would require the learner to define the floating points then explain the operation of adder and subtractor	CO 3
7	Explain shift micro-operations. Compare logical, arithmetic, and circular shifts with suitable examples and applications.	Understand	This would require the learner to recall number system then outline the complement operations	CO 3
8	Discuss the importance of micro-operations in processor organization and instruction execution.	Understand	This would require the learner to recall addressing mode then explain different addressing modes	CO 3
9	Explain the architecture and functions of the Control Unit with a neat block diagram.	Apply	This would require the learner to recall instructions then demonstrate the data transfer and manipulation instructions.	CO 3
10	Discuss the organization and operation of Control Memory in a microprogrammed control unit.	Understand	This would require the learner to recall floating then demonstrate the IEEE stand for binary floating-point arithmetic	CO 3
11	Explain address sequencing in detail. Discuss different methods of generating the next microinstruction address.	Understand	TThis would require the learner to recall multiplier then show the flow chart of multiplication process.	CO 3

12	Explain the format of a microinstruction and discuss its components with suitable examples.	Analyze	-	CO 3
13	Explain the execution of a microprogram using a suitable instruction example.	Understand	This would require the learner to define floating point then demonstrate the floating-point arithmetic	CO 3
14	Explain the design of a microprogrammed control unit with a neat diagram.	Understand	This would require the learner to recall multiplier then show the flow chart of multiplication process.	CO 3
15	Compare Hardwired Control Unit and Microprogrammed Control Unit based on architecture, speed, flexibility, and complexity.	Understand	This would require the learner to define arithmetic operation then explain the decimal arithmetic	CO 3
16	Explain the control signals generated during register transfer and memory transfer operations.	Understand	This would require the learner to recall memory then list the memory reference instructions.	CO 3
17	Discuss the role of microprogramming in implementing complex instruction sets.	Understand	This would require the learner to recall floating point operation then explain the multiplication procedure of floating-point numbers	CO 3
18	Explain the sequence of micro-operations required during instruction execution in a basic computer.	Apply	The learner to define number system then describes adder functions assigning to BCD adder.	CO 3
19	Discuss the advantages and disadvantages of microprogrammed control over hardwired control.	Understand	This would require the learner to recall multiplication then explain the algorithm	CO 3

20	Design a microprogrammed control unit capable of executing arithmetic, logic, and shift micro-operations. Explain its operation with neat diagrams.	Remember	-	CO 3
PART-C SHORT ANSWER QUESTIONS				
1	Define Register Transfer Language (RTL).	Remember	-	CO 3
2	What is Register Transfer?	Evaluate	This would require the learner to recall data then show representation of data	CO 2
3	Define Bus Transfer.	Remember	-	CO 3
4	What is Memory Transfer?	Understand	This would require the learner to recall number system then list number system	CO 3
5	Define Micro-operation.	Remember	-	CO 3
6	List the types of Micro-operations.	Remember	-	CO 3
7	Define Arithmetic Micro-operation.	Understand	This would require the learner to define registers then summarize the register addressing mode	CO 3
8	Define Logic Micro-operation.	Remember	-	CO 3
9	What is Shift Micro-operation?	Remember	-	CO 3
10	Differentiate between Logical Shift and Arithmetic Shift.	Understand	This would require the learner to recall transfer instructions then show the data transfer and manipulation	CO 3
11	What is Circular Shift?	Remember	-	CO 3
12	Define Control Unit.	Remember	-	CO 3

13	What is Control Memory?	Remember	-	CO 3
14	Define Microprogram.	Understand	This would require the learner to recall arithmetic operations then show the decimal functions.	CO 3
15	What is a Microinstruction?	Remember	-	CO 3
16	Define Address Sequencing.	Remember	-	CO 3
17	What is a Hardwired Control Unit?	Remember	-	CO 3
18	What is a Microprogrammed Control Unit?	Remember	-	CO 3
19	List the advantages of Register Transfer Language.	Remember	-	CO 3
20	State any four functions of the Control Unit.	Remember	-	CO 3

MODULE IV

CPU AND COMPUTER ARITHMETIC

PART A- PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS

1	Explain the instruction cycle with a neat flow diagram. Illustrate the Fetch, Decode, Execute, and Interrupt phases using an example instruction.	Understand	This would require the learner to recall the memory then explain the DMA transfer and show the block diagram.	CO 4
2	A processor executes the instruction ADD R1, M[2000H] . Explain the sequence of operations performed during the instruction cycle.	Understand	This would require the learner to recall the memory then explain the DMA I/O operation.	CO 4

3	Compare different data representation techniques used in CPU design. Analyze their impact on processor performance.	Understand	This would require the learner to recall the priority then explain the parallel priority	CO 4
4	Explain memory reference instructions. Demonstrate the execution of LOAD, STORE, ADD, and SUB instructions using suitable examples.	Understand	This would require the learner to recall the priority then explain the parallel priority	CO 8
5	A processor must transfer data from memory to the accumulator, perform addition, and store the result back in memory. Explain the execution sequence.	Understand	This would require the learner to recall the control memory then explain the data transfer with handshaking	CO 4
6	Explain the organization of Input–Output operations in a CPU. Discuss how the processor communicates with peripheral devices.	Understand	This would require the learner to recall the memory then explain the characteristics of memory	CO 4
7	A printer generates an interrupt while the CPU is executing a program. Explain how interrupt handling is performed.	Understand	This would require the learner to recall the instruction then explain the privileged instruction	CO 4
8	Compare maskable and non-maskable interrupts. Discuss their significance in processor operation.	Understand	This would require the learner to recall the bus then explain the asynchronous bus with read and write cycles	CO 4
9	Explain different addressing modes with suitable examples. Identify the effective address for each addressing mode.	Apply	The learner to define memory then explain about data transfer assigning to memory organization	CO 4

10	A program accesses memory using Immediate, Direct, Indirect, Register, and Indexed addressing modes. Compare their advantages and applications.	Understand	This would require the learner to recall the memory then explain the interface in memory organization	CO 4
PART-B LONG ANSWER QUESTIONS				
1	Explain CPU design with a neat block diagram. Discuss the functions of major CPU components.	Understand	This would require the learner to recall direct memory access capability and explain the I/O processor organization	CO 4
2	Explain the instruction cycle in detail. Describe each phase involved in instruction execution.	Understand	This would require the learner to recall different types of computer memories and compare between DRAM and RAM.	CO 4
3	Discuss different methods of data representation used in computer systems and explain their significance.	Understand	This would require the learner to recall the interfacing of I/O devices and compare types of interfacing	CO 4
4	Explain memory reference instructions with suitable examples and instruction execution sequences.	Understand	This would require the learner to recall control memory then explain the strobe control method and data transfer techniques	CO 6
5	Discuss Input–Output organization and interrupt processing in CPU design.	Understand	This would require the learner to recall the data transfer techniques then outline the asynchronous method.	CO 4
6	Explain the interrupt cycle. Describe interrupt handling using a neat flowchart.	Understand	This would require the learner to recall the interfacing then explain the input and output interfacings	CO 4

7	Explain different addressing modes with suitable examples. Compare their advantages and limitations.	Understand	This would require the learner to recall the memory then explain the memory mapping of direct and set associative	CO 4
8	Discuss Immediate, Direct, Indirect, Register, Indexed, Relative, and Base Register addressing modes.	Understand	This would require the learner to recall the interfacing of I/O devices and explain the bus and interface modules	CO 4
9	Explain data transfer instructions and discuss their role in CPU operation with suitable examples.	Understand	This would require the learner to recall I/O subsystem and demonstrate input interface circuit.	CO 4
10	Explain data manipulation instructions. Discuss arithmetic, logical, and shift operations performed by the CPU.	Understand	This would require the learner to recall different type of memories and compare between static and dynamic memories.	CO 4
11	Explain program control instructions and discuss branching, looping, subroutine calls, and returns.	Understand	TThis would require the learner to recall the basic structure of a memory hierarchy and relate the same with multiple levels of caches.	CO 4
12	Discuss CPU execution of arithmetic, logical, and control instructions with suitable examples.	Understand	This would require the learner to recall the interfacing of I/O devices and demonstrate single bus structures.	CO 4
13	Explain binary addition and subtraction using 2's complement representation with examples.	Understand	This would require the learner to recall the data transfer between sender and receiver and compare synchronous and asynchronous transmission	CO 4
14	Discuss overflow detection in binary arithmetic and explain methods to identify arithmetic overflow.	Understand	This would require the learner to recall the concept of interrupts and explain the masking of interrupts.	CO 4

15	Explain floating-point arithmetic operations. Discuss floating-point addition, subtraction, multiplication, and division.	Understand	This would require the learner to recall the basic structure of a memory hierarchy and explain the data transfer between multiple levels of Memories.	CO 4
16	Explain the IEEE-754 floating-point format and its role in floating-point arithmetic.	Understand	This would require the learner to recall the memory then explain the kinds of memory function.	CO 4
17	Explain the architecture and operation of the Decimal Arithmetic Unit with a neat block diagram.	Apply	The learner to define memory then describe about mapping assigning to functions of cache	CO 4
18	Compare Binary Arithmetic Unit and Decimal Arithmetic Unit with suitable examples.	Understand	This would require the learner to recall the memory then explain the kinds of memory function.	CO 4
19	Explain the role of CPU design in improving processor performance and instruction execution efficiency.	Understand	This would require the learner to recall the memory then explain the DRAM configuration	CO 4
20	Design a simple CPU capable of executing arithmetic, memory reference, I/O, interrupt, and program control instructions. Explain its operation with neat diagrams.	Understand	This would require the learner to recall the memory then explain the paging and segmentation	CO 4
PART-C SHORT ANSWER QUESTIONS				
1	Define CPU Design.	Remember	-	CO 4
2	What is an Instruction Cycle?	Remember	-	CO 4
3	List the phases of the Instruction Cycle.	Remember	-	CO 4
4	Define Memory Reference Instruction.	Remember	-	CO 4
5	What is Data Representation?	Remember	-	CO 4
6	Define Input–Output Operation.	Remember	-	CO 4
7	What is an Interrupt?	Remember	-	CO 4

8	Differentiate between Hardware Interrupt and Software Interrupt.	Remember	-	CO 4
9	Define Addressing Mode.	Remember	-	CO 4
10	List different Addressing Modes.	Remember	-	CO 4
11	What is a Data Transfer Instruction?	Remember	-	CO 4
12	Define Data Manipulation Instruction.	Remember	-	CO 4
13	What is a Program Control Instruction?	Remember	-	CO 4
14	Define Binary Addition.	Remember	-	CO 4
15	What is 2's Complement Subtraction?	Remember	-	CO 4

16	Define Floating Point Arithmetic.	Remember	-	CO 4
17	What is IEEE-754 Standard?	Remember	-	CO 4
18	Define Decimal Arithmetic Unit.	Remember	-	CO 4
MODULE V				
INPUT–OUTPUT AND MEMORY ORGANIZATION				
PART A-PROBLEM SOLVING AND CRITICAL THINKING QUESTIONS				
1	Explain the concept of memory hierarchy. Analyze how different levels of memory improve the overall performance of a computer system.	Understand	This would require the learner explains the theory of parallelism and interprets its real-world implications in hardware systems.	CO 5
2	A processor frequently accesses the same data. Explain how cache memory reduces memory access time and improves execution speed.	Understand	This would require the learner interprets pipeline architecture and explains its impact on performance along with associated hazards.	CO 5
3	Compare SRAM and DRAM used in main memory. Analyze their suitability for cache memory and primary memory applications.	Understand	This would require the learner compares two types of pipelines and explains each with appropriate examples. .	CO 5
4	Explain Associative Memory. Discuss its architecture, operation, advantages, and applications	Understand	This would require the learner explains vector processing concepts and contrasts it with scalar processing using examples. .	CO 5
5	Explain the organization of main memory. Illustrate memory read and memory write operations with suitable diagrams.	Understand	This would require the learner interprets the structure and function of array processors with illustrative diagrams.	CO 5
6	Compare main memory and auxiliary memory. Explain their roles in data storage and program execution.	Understand	This would require the learner recalls the features of multiprocessors and explains key differences from uniprocessors.	CO 5
7	Explain associative memory and discuss how it differs from conventional memory organization.	Understand	This would require the learner explains interconnection architectures and compares their design trade-offs.	CO 5

8	A cache memory has a hit ratio of 95%. Analyze how cache hits and cache misses affect processor performance.	Understand	This would require the learner interprets arbitration protocols and explains how resource access is coordinated across processors.	CO 5
PART-B LONG ANSWER QUESTIONS				
1	Explain the organization of the memory hierarchy with a neat diagram. Discuss the characteristics of each level of memory.	Understand	This would require the learner comprehends and explains the real-world architectural and performance benefits of multiprocessors, interpreting how these improve system throughput, fault tolerance, and scalability using examples.	CO 5
2	Explain the architecture and organization of main memory. Discuss memory read and write operations.	Understand	This would require the learner breaks down both SMP and AMP, categorizes their architectural differences, and contrasts their operational characteristics, showing analytical depth.	CO 5
3	Compare Main Memory and Auxiliary Memory with suitable examples and applications.	Understand	This would require the learner demonstrates understanding of processor evolution by explaining the progression to multicore architectures and its impact on parallelism and efficiency.	CO 5
4	Explain Associative Memory. Discuss its architecture, operation, advantages, and applications.	Understand	This would require the learner deconstructs complex hardware challenges in multicore environments, explaining causes and effects on performance to show analytical thinking.	CO 6
5	Explain Cache Memory organization. Discuss cache mapping techniques and cache replacement policies	Understand	This would require the learner evaluates inefficiencies at the software layer, critiques current programming practices, and proposes justified solutions using optimized compilation and parallel logic.	CO 5

6	Explain cache hit, cache miss, and hit ratio. Discuss their impact on system performance.	Apply	This would require the learner interprets and visualizes architecture using diagrams, identifying how communication and resource sharing occur between cores.	CO 5
7	Explain Virtual Memory organization. Discuss paging, address translation, and page replacement techniques.	Understand	This would require the learner dissects threading models, compares SMT and hyper-threading techniques, and analyzes their effects on CPU utilization.	CO 6
8	Compare Cache Memory and Virtual Memory with suitable examples.	Understand	This would require the learner explains a commercial processor's technical architecture, recognizing and interpreting key hardware metrics.	CO 6
9	Explain instruction pipelining. Discuss the stages of a pipeline with suitable diagrams.	Analyze	This would require the learner compares technical specifications and evaluates progress in architectural design and performance.	CO 6
10	Explain pipeline hazards and discuss techniques used to overcome them.	Apply	This would require the learner develops a benchmarking scenario to compare processor performance through applied example.	CO 6
11	Explain the organization of the Input–Output Interface with a neat block diagram.	Understand	This would require the learner applies Amdahl's Law formula to a scenario to calculate performance limits in parallel processing.	CO 5
12	Discuss asynchronous data transfer. Explain handshaking and strobe control methods.	Understand	This would require the learner interprets OS-level scheduling concepts and explains how cores are balanced during task execution.	CO 5
13	Explain different modes of data transfer. Compare Programmed I/O, Interrupt-driven I/O, and Direct Memory Access.	Understand	This would require the learner explains ILP techniques and interprets architectural strategies used for parallel instruction execution.	CO 6

14	Explain Priority Interrupt. Discuss interrupt priority schemes and interrupt handling procedures.	Understand	This would require the learner identifies pipeline hazards and explains how they can be mitigated in modern CPUs. .	CO 6
15	Explain the architecture and operation of Direct Memory Access (DMA) with a neat block diagram.	Understand	This would require the learner creates a pipeline model and computes performance improvements with illustrative timing.	CO 6
16	Compare Interrupt-driven I/O and DMA. Discuss their advantages and limitations.	Analyze	This would require the learner describes Flynn's model and classifies computer architectures based on parallelism.	CO 5
PART-C SHORT ANSWER QUESTIONS				
1	Define Memory Hierarchy.	Remember	—	CO 5
2	What is Main Memory?	Remember	—	CO 5
3	Define Auxiliary Memory.	Remember	—	CO 5
4	What is Associative Memory?	Remember	—	CO 5
5	Define Cache Memory.	Remember	—	CO 5
6	What is a Cache Hit?	Remember	—	CO 5
7	Define Virtual Memory.	Remember	—	CO 5
8	What is Paging?	Remember	—	CO 5
9	Define Pipelining.	Remember	—	CO 5
10	List the stages of Instruction Pipelining.	Remember	—	CO 5
11	Define Input–Output Interface.	Remember	—	CO 5

12	What is Asynchronous Data Transfer?	Remember	—	CO 5
13	Define Handshaking.	Remember	—	CO 5
14	What is Programmed I/O?	Remember	—	CO 5
15	Define Interrupt-driven I/O.	Remember	—	CO 5

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